

Power Budget for HMI - Jacob Alger							
Team Number:	Team 202						
Project Name:	Cropscout						
Team Member Names:	Jacob Alger, Caleb Yuen, Aaron Kiem, Isaiah Johnston, Cristopher Gutierrez, Asadbek Ruziev, Tyler Dean						
Version:	1						
A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements							
All Major Components	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	ESP32 Microcontroller	ESP32-S3-WROOM-1-N4	-0.3V to 3.6V	1	335	335	mA
	OLED Screen	GME12864-17	-0.3V to 4V	1	40	40	mA
B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.							
+3.3V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	ESP32 Microcontroller	ESP32-S3-WROOM-1-N4	-0.3V to 3.6V	1	335	335	mA
	OLED Screen	GME12864-17	-0.3V to 4V	1	40	40	mA
					Subtotal	375	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	468.75	mA
c1. Regulator or Source Choice	3.3V Switching Regulator	LM2575-3.3WU-TR	4V to 40V	1	1000	1000	mA
					Total Remaining Current Available on 3.3V Rail	531.25	mA

+9V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	3.3V Switching Regulator	LM2575-3.3WU-TR	4V to 40V	1	1000	1000	mA
	Subtotal					1000	mA
	Safety Margin					25%	
	Total Current Required on +9V Rail					1250	mA
c2. Regulator or Source Choice	DC Power Jack	PJ-102AH	24V	1	5000	5000	mA
	Total Remaining Current Available on +9V Rail					3750	mA
C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above							
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need							
External Power Source 1	Component Name	Part Number	SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	B09ZTKTLGW	110VAC	12V	15000	15000	mA
Power Rails Connected to External Power Source 1	DC Power Jack	PJ-102AH	24V	9V	5000	5000	mA
	3.3V Switching Regulator	LM2575-3.3WU-TR	4V - 40V	1	1000	1000	mA
	Total Remaining Current Available on External Power Source 1					9000	mA